



# **Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques**

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## **DECLARATION AND APPROVAL**

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I declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of my knowledge and belief, the proposal contains no material previously published or written by another person except where due reference is made in the proposal itself.

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### **APPROVAL**

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## ABSTRACT

Customer churn prediction in the telecommunications sector is a persistent challenge that significantly affects profitability, competitiveness, and long-term sustainability. The study begins by outlining the background and research problem, noting the increasing competition in the sector and the limitations of traditional churn management approaches. It highlights the objectives of identifying key churn predictors, developing a machine learning–based prediction model, evaluating multiple algorithms, and designing a deployable dashboard to support decision-making.

A review of relevant literature synthesizes theoretical frameworks such as Customer Relationship Management (CRM) and Technology Adoption Theory, along with recent empirical studies on churn prediction. Prior work demonstrates the potential of machine learning and deep learning models but also reveals gaps related to interpretability, profit-sensitive evaluation, and limited focus on other telecommunications markets.

The methodology follows the CRISP-DM framework and utilizes secondary data from the Iranian Churn (UCI) and Duke Cell2Cell (Teradata) datasets. Data preprocessing steps include cleaning, feature engineering, class imbalance handling, and encoding. Several models, including Logistic Regression, Decision Trees, Random Forest, and LightGBM, will be compared, with hyperparameter tuning and interpretability techniques such as SHAP applied to enhance performance and stakeholder trust.

Expected outcomes include the development of a robust churn prediction system with high predictive accuracy, interpretability, and scalability. From an operational perspective, the system will enable proactive customer retention strategies, optimized resource allocation, and continuous monitoring. This research contributes a deployable and replicable framework that addresses the immediate business need for churn reduction while offering methodological insights applicable to other telecommunications markets.

**Keywords:** Random Forest, Customer Churn, Telecommunication, Customer retention

## **ACRONYMS**

1. ANNs – Artificial Neural Networks
2. CNNs – Convolutional Neural Networks
3. CRM- Customer Relationship Management
4. DL – Deep Learning
5. DT – Decision Tree
6. ERT – Extra Random Trees
7. FT – Functional Trees
8. GBM – Gradient Boosting
9. k-NN – k-Nearest Neighbors
10. LMT – Logistic Model Tree
11. LR – Logistic Regression
12. ML – Machine Learning
13. RF – Random Forest

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# 1. CHAPTER 1: INTRODUCTION

## 1.1 Background

Customer churn defined as customers discontinuing their subscription or services in favor of competing offers represents a critical and persistent challenge within the global telecommunications industry. Churn profoundly affects business profitability, operational sustainability, and long-term market competitiveness. Globally, annual churn rates in telecommunications range between 20% and 40%, contributing to substantial revenue losses for service providers each year and underscoring the urgency for proactive prediction and retention strategies, *Annual Churn Rate in Global Telecommunications Sector* (2023).

From an economic perspective, customer retention is significantly more cost-effective than acquisition. Studies indicate that acquiring a new customer can be five to seven times more expensive than retaining an existing one Kotler et al. (2021). High churn rates can cause inflate acquisition costs, erode market share, reduce revenue stability, and constrain organizational competitiveness. Socially and culturally, stable telecommunication access is essential for enabling critical services, supporting digital inclusion, and facilitating economic participation, particularly in emerging economies such as those in Sub-Saharan Africa *Mobile Economy Report: Sub-Saharan Africa 2022* (2022).

Despite ongoing investment in churn reduction strategies, many telecommunications companies still rely on traditional methods of churn analysis and prediction. These methods often use static and retrospective datasets that inadequately capture dynamic shifts in customer preferences, external socio-economic influences, and seasonal or event-driven fluctuations in usage Chang et al. (2024). Such approaches frequently lead to subjective decision-making, inefficient allocation of retention resources, and limited predictive accuracy.

Recent advances in data science, particularly in Machine Learning (ML), present an opportunity to address these limitations. Predictive analytics, leveraging algorithms such as Logistic Regression, Decision Trees, and Random Forest Classifiers, have demonstrated strong potential for modeling complex customer behaviors and identifying at-risk subscribers before they leave Rahman et al. (2024). These models can integrate large-scale, heterogeneous datasets, uncover non-linear patterns, and generate actionable insights with higher accuracy than conventional approaches.

Telecommunications market is characterized by high competition, rapid digital adoption, and socio-economic reliance on mobile services, these analytical capabilities are particularly relevant. By

combining behavioral, demographic, and service usage data with advanced ML techniques, providers can proactively intervene to retain customers, optimize marketing expenditure, and strengthen customer loyalty.

## **1.2 Research Problem**

The telecommunications sector faces intensifying competition, evolving consumer expectations, and increased vulnerability to churn due to competitive offers and service quality perceptions. Current churn prediction and management approaches in many organizations remain reactive, relying on static indicators and manual analyses that are time-consuming, computationally inefficient, and prone to human bias. These methods struggle to capture real-time changes in customer behavior and broader socio-economic shifts, resulting in suboptimal retention strategies and high revenue leakage Mwaura (2021).

## **1.3 Research Objectives**

This study seeks to address the identified gaps through the following objectives:

### **Main Objective**

To identify key churn predictors in telecommunication industry.

### **Specific Objectives**

- i. To describe the challenges and limitations of existing churn prediction approaches in the telecommunications industry.
- ii. To develop a machine learning based churn prediction model using telecommunications data.
- iii. To evaluate and compare the performance of Machine Learning algorithms.
- iv. Deploy Machine learning models in an interactive dashboard.

## **1.4 Significance and Justification**

From a global perspective, effective churn management is critical for sustaining profitability and competitiveness in the telecommunications sector. Retention-focused strategies reduce the high costs associated with customer acquisition as well as stabilize revenue streams, safeguard market share, and enhance long-term customer value.

In emerging markets, the implications of churn extend beyond corporate profitability. Telecommunications services are deeply embedded in daily life, acting as enablers for financial transactions, e-learning, e-health, and small business operations. High churn rates risk excluding segments of the population from these benefits, slowing progress toward digital inclusion and socio-economic development.

This research is justified on three key fronts:

- **Economic efficiency:** By identifying at-risk customers before they leave, providers can target retention efforts more precisely, thereby reducing costs and maximizing return on marketing investment.
- **Operational effectiveness:** The integration of the predictive model into CRM systems ensures actionable insights are delivered to decision-makers in real time, enabling timely and effective interventions.
- **Scalability and replicability:** While tailored to Kenya's socio-economic and competitive environment, the proposed framework offers a methodological blueprint that can be adapted for other markets and industries facing similar churn-related challenges.

Ultimately, this study contributes to both the academic body of knowledge on churn prediction and the practical toolkit available to telecommunications providers.

By leveraging advanced ML techniques and embedding them into operational workflows, the research aims to create a sustainable, data-driven approach to customer retention in highly competitive markets.

#### **1.4.1 Research Questions**

To address the stated objectives, the study is guided by the following research questions:

1. What challenges and limitations exist in current churn prediction approaches within the telecommunications industry?
2. Which demographic, contractual, financial, and usage related factors are the most significant predictors of customer churn?
3. How can machine learning models be developed and optimized to predict churn in the telecommunications sector?



4. How does the performance of different ML algorithms (e.g., Logistic Regression, Random Forest, LightGBM) compare in predicting customer churn?
5. How can a predictive churn model be integrated into a CRM system through an interactive dashboard to support real-time decision-making and interventions?

## **2. CHAPTER 2: LITERATURE REVIEW**

The telecommunications industry has undergone rapid transformation in recent years due to technological advancement, market liberalization, and shifting consumer expectations. While these trends have enhanced access and service delivery, they have also intensified competition, making customer retention a strategic priority. Customer churn, the phenomenon where subscribers discontinue their relationship with a provider, poses one of the most pressing challenges. Globally, annual churn rates in telecommunication range between 20–40%, resulting in substantial financial losses for service providers and reinforcing the importance of predictive approaches Khan et al. (2024).

### **2.1 Theoretical Framework**

Customer churn has long been viewed as both an economic and relational problem. As noted by Smith et al. (2021), modern service industries face churn risks driven not only by service quality but also by customer experience and digital interactions. Rogers' Diffusion of Innovation theory remains relevant, explaining why customers adopt, continue, or abandon telecom services depending on perceived advantage and compatibility. Likewise, Customer Relationship Management (CRM) frameworks emphasize proactive retention through personalization, loyalty programs, and predictive insights. These theories collectively highlight the multidimensional nature of churn, spanning economic, behavioral, and technological perspectives.

### **2.2 Empirical Studies**

Recent years have witnessed significant advances in churn modeling using machine learning and deep learning. For instance, Usman-Hamza et al. (2022) demonstrated that ensemble decision forests with weighted voting strategies outperformed baseline classifiers in imbalanced telecom churn datasets. Similarly, Saha et al. (2023) highlighted the superiority of convolutional neural networks (CNNs) in hybrid datasets from India and the U.S., emphasizing that ROI-sensitive evaluation matters as much as predictive accuracy.

Context-specific models remain essential. In Nepal, Nagarkar (2022) applied XGBoost to 52,332 telecom customers, showing that recharge frequency and inactivity duration are strong predictors of churn, enabling targeted retention campaigns. In Syria, integrated social network analysis (SNA) features, improving AUC from 0.84 to 0.933, demonstrating the value of relational data. More recently, Poudel and Sharma (2024) introduced explainer-aware churn models that combined interpretability

methods with predictive algorithms, showing the importance of stakeholder trust in adopting churn systems.

At a broader level, Shahabikargar and Kiani (2025) reviewed churn prediction across industries and highlighted the growing importance of integrating external and real-time data, as well as the need to balance performance with transparency. Their findings reinforce that churn solutions must be both accurate and interpretable to be actionable in practice.

## **2.3 Customer Churn in the Telecommunications Industry**

Customer churn remains a major challenge for the telecom industry, reducing revenue stability and profitability. Studies confirm that global churn rates can exceed 30%, even in mature markets, highlighting the urgency for robust retention strategies Khan et al. (2024). In Sub-Saharan Africa, where telecoms are a lifeline for financial transactions, e-learning, and small business operations, the stakes are particularly high Mwaura (2021). While predictive modeling has advanced significantly, few studies focus directly on African contexts, revealing a critical gap in geographic representation.

## **2.4 Industry Relevance and Contribution**

This review highlights key insights:

Ensemble and deep learning approaches consistently outperform traditional models. Interpretability and business ROI are increasingly important evaluation criteria, and other markets remain underexplored.

The present study contributes by applying comparative machine learning analysis to the Iranian Churn dataset and the Duke Cell2Cell dataset, thereby addressing gaps in cross-market generalization, interpretability, and profit-sensitive evaluation.

The proposed framework provides a scalable and practical approach tailored to the realities of telecommunication churn management in emerging markets.

## **2.5 Gaps in Existing Literature**

- **Feature Limitations:** Available datasets mainly include structured CRM variables (billing, tenure, usage), but exclude important signals such as network quality, digital engagement, and promotions. This narrows the feature space for building robust churn models Mwaura (2021).

- **Evaluation Metrics Bias:** Existing research often prioritizes accuracy and AUC, while overlooking profit-sensitive or ROI-based metrics Shahabikargar and Kiani (2025). This creates a gap between academic results and actionable business insights.
- **Interpretability Challenges:** High-performing ensemble and deep learning models are frequently treated as “black boxes.” Limited interpretability makes it difficult for managers to trust model outputs Khan et al. (2024) and translate them into real retention strategies.

## 2.6 Conceptual Framework

The conceptual framework for this study comprises the following components:

- **User Behavior Analysis:** Examination of customer demographics, usage patterns, contractual terms, and payment behaviors to identify churn predictors.
- **Machine Learning Models:** Implementation of algorithms including Logistic Regression, Random Forest, Decision Trees, and Gradient Boosting to predict churn, with comparative performance analysis.
- **User Retention Strategies:** Translation of model outputs into actionable retention initiatives tailored to the Kenyan telecommunications market.
- **Performance Evaluation:** Use of metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to assess predictive performance.
- **Contextual Considerations:** Incorporation of Kenya’s socio-economic and operational realities to ensure recommendations are practical and market-relevant.

### **3. CHAPTER 3: METHODOLOGY**

This study adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology to ensure a structured, repeatable, and scalable approach to customer churn prediction in the Kenyan telecommunications sector.

#### **3.1 Data Understanding**

This study will employ secondary datasets widely used in churn prediction research: the Iranian Churn dataset (UCI Repository) and the Duke Cell2Cell dataset (Teradata Center, Duke University). The Iranian dataset captures attributes such as customer demographics, subscription type, and service usage patterns, while the Duke Cell2Cell dataset includes contractual information, billing records, and behavioral variables. Together, these datasets provide a comprehensive representation of churn dynamics across different telecommunication markets.

A preliminary exploratory data analysis (EDA) will be performed to characterize churn and non-churn distributions, identify potential class imbalances, and examine variable relationships. Descriptive statistics and visualization techniques such as histograms, scatter plots, and correlation matrices will be used to highlight customer usage behaviors, billing trends, and contractual drivers of churn. This step ensures a robust understanding of the datasets prior to model building and aligns with best practices in churn research, where exploratory analysis plays a critical role in feature engineering and data-driven model selection Wagh et al. (2024).

#### **3.2 Data Collection**

This study will utilize secondary data obtained from two well-established telecommunications churn datasets: the Iranian Churn dataset, hosted by the UCI Machine Learning Repository, and the Duke Cell2Cell dataset, available through the Teradata Center for Customer Relationship Management at Duke University. The Iranian dataset provides customer-level attributes such as demographics, subscription types, and service usage, while the Duke Cell2Cell dataset offers extensive behavioral, contractual, and financial information on telecommunication customers.

The use of these datasets ensures access to large-scale, high-quality, and pre-validated records that capture diverse aspects of customer engagement and churn behavior. Importantly, both datasets allow for comparative analysis across different market contexts, strengthening the generalizability of findings. All data are fully anonymized and publicly available, ensuring compliance with ethical

standards and data governance requirements. By leveraging these datasets, the study gains a robust foundation for developing, testing, and validating machine learning models for churn prediction, while addressing existing gaps in literature related to transferability and cross-market benchmarking.

### **3.3 Data Cleaning**

Data cleaning will constitute a critical component of the data pre-processing phase, undertaken to enhance the quality, accuracy, and reliability of the dataset prior to analysis. Given that the data will be obtained through structured questionnaires, potential sources of error may include typographical mistakes, incomplete responses, inaccurately recorded values, and inconsistencies in categorical labels or formatting.

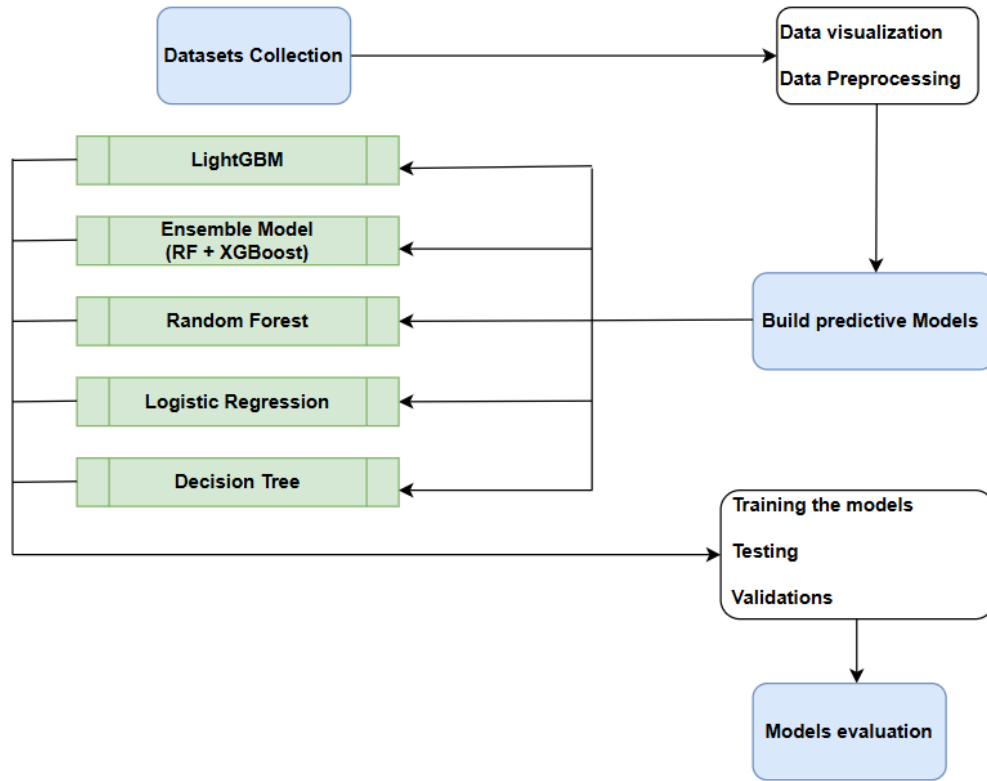
A systematic cleaning process will be applied to address these issues. Specifically, missing values will be imputed using appropriate statistical or rule-based methods to preserve the representativeness and distributional characteristics of the data. Erroneous entries and illogical values will be identified and corrected through consistency checks and cross-validation against predefined logical rules. Furthermore, categorical and numerical variables will be standardized to ensure uniformity across all responses, thereby facilitating compatibility with subsequent analytical procedures.

This rigorous data cleaning protocol is intended to minimize noise, reduce the risk of bias, and ensure that the final dataset is robust, coherent, and fit for reliable statistical analysis and predictive modelling.

## **3.4 Machine Learning Modelling**

### **3.4.1 Proposed Workflow**

The proposed workflow for this study is illustrated in Figure 3.1. It outlines the systematic process of building and evaluating machine learning models for customer churn prediction.



**Figure 3.1: Proposed Workflow**

### 3.4.2 Data Exploration

Through data exploration, this study will derive valuable insights into the distribution, variability, and structure of the collected dataset. This phase will involve performing descriptive statistical analyses, generating visualizations, and computing summary statistics to identify patterns, trends, and potential relationships within the data. Exploratory Data Analysis (EDA) techniques, including histograms, scatter plots, box plots, and correlation matrices, will be used to assess the relationships between customer demographic factors, contractual attributes, financial metrics, and service usage patterns. These insights will inform the selection of appropriate machine learning models and preprocessing strategies.

### 3.4.3 Treating Missing Data

Addressing missing data is critical to ensuring the completeness and quality of the dataset. Missing values will be handled based on their nature and extent. Where feasible, imputation techniques such as mean imputation, regression-based imputation, or domain-informed replacement will be applied. Where the proportion of missing data is excessive or the missingness is deemed non-random, the

affected entries may be excluded to prevent bias. The choice of imputation method will be guided by the variable type, data distribution, and potential impact on predictive performance.

#### **3.4.4 Treating Outliers**

Outliers have the potential to distort statistical summaries and reduce the accuracy of predictive models. Outlier detection will be conducted using statistical thresholds (e.g., z-scores, interquartile ranges) and visual diagnostics (e.g., box plots, scatter plots). Detected outliers will be evaluated in consultation with subject matter experts to determine whether they represent valid extreme cases or erroneous entries. Appropriate treatments, such as capping, transformation, or removal, will be applied as necessary.

#### **3.4.5 Data Type Conversion**

Data type conversion will ensure that variables are consistently represented for analysis. Categorical variables will be converted into standardized label formats, and numerical variables will be cast to appropriate numeric data types. This process ensures compatibility with machine learning algorithms and facilitates reproducibility.

#### **3.4.6 Data Transformation**

##### **Feature Engineering**

The dataset will be refined to improve model predictive capacity through feature engineering. This process will involve generating new variables or modifying existing ones to capture latent patterns in customer churn behavior. Examples may include the computation of customer tenure categories, normalized spending ratios, and derived usage intensity scores.

##### **Feature Scaling**

Feature scaling will standardize the range of numerical variables to ensure equitable influence on model training. Standardization using z-score normalization and normalization using Min-Max scaling will be considered. These methods prevent variables with large numerical ranges (e.g., monthly charges) from dominating the learning process.

##### **Feature Selection**

Relevant features will be selected to reduce dimensionality, mitigate overfitting, and improve interpretability. Statistical tests, correlation analysis, and model-based feature importance rankings will



guide the selection process. Irrelevant or redundant features will be excluded to enhance computational efficiency.

## **Data Encoding**

Categorical attributes will be transformed into numerical representations suitable for machine learning models. One-hot encoding, label encoding, or binary encoding will be applied depending on the nature of the variable and the target algorithm's requirements.

### **3.4.7 Model Selection**

Customer churn prediction constitutes a binary classification task, requiring the accurate categorization of customers into churn and non-churn groups based on their demographic, contractual, financial, and behavioral attributes. To address this challenge, the study will adopt a comparative modelling approach that evaluates several machine learning algorithms, each chosen for its unique strengths and suitability for the churn prediction problem in the telecommunications sector.

Logistic Regression will be employed as a foundational benchmark model. Its simplicity, computational efficiency, and ease of interpretation make it a reliable baseline for classification tasks. By estimating churn probabilities using a linear combination of input features, LR provides a clear understanding of how each predictor influences churn likelihood, which is valuable for decision-making and communication with non-technical stakeholders.

Random Forest, an ensemble method that constructs multiple decision trees and aggregates their predictions, will be used for its ability to handle large, heterogeneous datasets while maintaining high predictive performance. RF reduces the risk of overfitting, provides inherent measures of feature importance, and is robust to noisy data—qualities that are crucial when working with real-world telecommunications datasets.

Decision Tree models will be included for their interpretability and straightforward visualization of decision rules. DTs recursively split the dataset into smaller, more homogeneous groups based on feature values, allowing for intuitive mapping of customer attributes to churn outcomes. This transparency supports stakeholder trust and facilitates actionable insight generation.

Light Gradient Boosting Machine will be utilized for its state of the art performance and computational efficiency. LightGBM is optimized for large scale and high dimensional data, making it highly suitable for churn prediction in contexts where the number of features and records is substantial. Its gradient boosting framework enables the capture of complex, non-linear relationships between customer characteristics and churn behavior.

An ensemble model will be developed to combine the strengths of multiple base learners, enhancing robustness and stability in predictions. By aggregating the outputs of diverse algorithms, the ensemble approach is expected to reduce individual model biases and variances, thereby improving overall generalization performance.

Finally, a Logistic Regression model trained on data augmented using the Synthetic Minority Oversampling Technique (SMOTE) will be implemented to specifically address class imbalance. Churn datasets often contain disproportionately fewer churners compared to non-churners, and SMOTE will synthetically generate new minority class samples within the training set. This approach is expected to improve the model's sensitivity in detecting churners, which is essential for proactive retention strategies.

Through this multi-model strategy, the study will ensure a rigorous evaluation of algorithmic performance, leading to the selection of a final model that optimally balances predictive accuracy, interpretability, and operational applicability within the telecommunications context.

### **3.4.8 Model Evaluation**

The models will be trained using an 80/20 train-test split with stratified sampling to preserve churn class proportions. Performance will be evaluated using:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) will be computed to measure the model's discriminatory ability. Given the business cost of false negatives in churn prediction, recall will be prioritized during model ranking.

## **3.5 Model Deployment**

The selected model will be deployed via an Application Programming Interface (API) to enable integration with existing customer management systems. Deployment through an API will provide modularity, maintainability, and scalability, allowing the predictive engine to serve multiple client applications concurrently.

**Deployment Framework:** FastAPI will be used as the deployment framework due to its speed, built-in data validation, automatic documentation, and asynchronous capabilities. The deployed model will return churn probability scores and key predictive factors for each customer record. These outputs will support targeted retention campaigns and ongoing monitoring of churn risk.

## 4. CHAPTER 4: EXPECTED OUTCOMES

The proposed study is expected to generate a robust, data-driven decision-support framework for customer churn prediction in the telecommunications sector. By leveraging secondary datasets from the Iranian Churn (UCI) and Duke Cell2Cell (Teradata Center) repositories, the research will provide valuable insights into churn behavior that can be adapted to emerging markets such as Kenya. The outcomes will address existing research gaps in cross-market transferability, interpretability, and profit-sensitive evaluation.

From a technical perspective, the final system will deliver:

- High predictive performance across recall, precision, F1-score, and ROC-AUC, ensuring that churners are identified accurately and with minimal delay.
- Interpretable model outputs through explainable AI techniques such as SHAP values, enabling business stakeholders to understand and act upon the most influential churn drivers.
- A scalable and modular architecture suitable for integration with Customer Relationship Management (CRM) platforms, capable of handling large volumes of customer records efficiently.

From an operational perspective, the system will enable:

- Proactive retention strategies that reduce revenue leakage by identifying high-risk customers before service termination occurs.
- Improved allocation of marketing and customer care resources by prioritizing interventions for high-value and high-risk subscribers.
- Continuous performance monitoring and model updating, with automated recalculation of evaluation metrics and feature-importance tracking to ensure model relevance in changing market conditions.

This research will contribute both a deployable, real-world churn prediction solution and a methodological framework that is replicable in other African telecommunications markets. By combining predictive accuracy with interpretability and business alignment, the study will support sustainable customer retention strategies, enhance operational efficiency, and promote digital inclusion through improved service continuity.

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